

Large Intestine: Diverticular Disease, Haemorrhoids and Appendicitis

Diverticular disease (connective tissue and colon muscle), haemorrhoids, appendicitis and bowel transit (irritable bowel syndrome)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The predisposition to appendicitis is low: the three genomic scores are at percentiles 15.0, 5.2 and 30.7 (**consensus 15.0**). The same holds for haemorrhoids: the consensus of the scores with good coverage is **18.9** (2022 genomic score at percentile 8.5) and the loci burden is at percentile 19.3, in line with the 29th percentile in the Coagulation, Thrombosis and Veins report; SMAD3, CDKN2B-AS1 and XKR9 carry no risk alleles. Irritable bowel syndrome has a loci burden at percentile 24.0, and CADM2 lacks the risk allele.

Unfavourable. Diverticular disease is the point to watch. The burden of 214 loci is at percentile **97.8**, the score with 23,800 variants at percentile 85.5, and the consensus of the 5 scores at **68.7** (the most recent genomic score, from 2024, at percentile 58.9). In the genes: **GPR158** T/T (2 alleles) and 1 allele each in **ARHGAP15** (the largest locus), COLQ, EFEMP1 (fibulin-3) and COL6A1 (collagen VI); elastin (ELN) and CALCB carry no risk alleles. For haemorrhoids there are isolated alleles in HMGA2, SFMBT1 and MUC12, with no overall effect.

No judgement. Stool frequency has genetics for faster transit (loci burden at percentile 99.2), which mechanically favours a lower risk of diverticular disease.

What it means. The large intestine has a mixed profile: appendicitis and haemorrhoids are unlikely, but there is a moderately high predisposition to diverticular disease, which is mainly a disease of the connective tissue and muscle of the colon wall. Diverticula become very common with the years and are almost always silent; what prevents them most is high fibre, physical activity, not smoking and avoiding chronic NSAID use. High fibre is already in the plan.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (statistical imputation, public panel of 30x whole genomes) with quality DR2 $\geq$ 0.8. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 17 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 8 from the chip, 9 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 is palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 12 PGS Catalog scores calculated locally with 5,725,774 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population (n = 633), calculated with exactly the same variants. Percentile $\geq$ 80: MORE LIKELY/HIGH; $\leq$ 20: LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population (n = 633) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 310 loci, 289 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Diverticular disease	68.7	5 of 5	58.9	3
Haemorrhoids	18.9	2 of 4	8.5	1
Appendicitis	15.0	3 of 3	15.0	3

In all these scores, a higher percentile means more risk. Two haemorrhoid scores (2021 and 2024) have 45–57% coverage and are left out of the consensus.

Diverticular disease

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Diverticular disease (genomic, 2024)	PGS004614 (Wu 2023, 1.08 million variants)	1,005,374 of 1,082,282 (coverage 92%)	58.9	TYPICAL	typical
Diverticular disease of the intestine (genomic)	PGS004474 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	68.7	TYPICAL	typical
Diverticulosis (genomic)	PGS002068 (Privé 2022, 831,000 variants)	815,229 of 831,212 (coverage 97%)	48.8	TYPICAL	typical
Diverticulosis	PGS001857 (Privé 2022, 24,000 variants)	23,066 of 23,800 (coverage 97%)	85.5	MORE LIKELY	unfavourable
Diverticular disease and diverticulitis	PGS000996 (PGS Catalog 2021, 368 variants)	308 of 368 (coverage 83%)	71.6	TYPICAL	typical

Haemorrhoids

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Haemorrhoids (genomic, 2024)	PGS004933 (Moreno-Grau 2024, 185,000 variants)	131,363 of 184,519 (coverage 57%)	71.4	TYPICAL	typical
Haemorrhoids (genomic)	PGS002058 (Privé 2022, 780,000 variants)	765,155 of 780,418 (coverage 97%)	8.5	LESS LIKELY	favourable
Haemorrhoids	PGS001846 (Privé 2022, 5,434 variants)	5,317 of 5,434 (coverage 97%)	29.2	TYPICAL	typical
Haemorrhoids	PGS001024 (Tanigawa 2022, 786 variants)	341 of 784 (coverage 45%)	84.4	MORE LIKELY	unfavourable

Appendicitis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Appendicitis (genomic, 2024)	PGS004876 (Jermy 2024, 602,000 variants)	582,355 of 601,835 (coverage 96%)	15.0	LESS LIKELY	favourable
Acute appendicitis (genomic)	PGS001785 (Wang 2023, 911,000 variants)	884,989 of 911,334 (coverage 96%)	5.2	LESS LIKELY	favourable
Diseases of the appendix (genomic)	PGS002064 (Privé 2022, 497,000 variants)	486,638 of 497,422 (coverage 97%)	30.7	TYPICAL	typical

4. Key variants by topic

Replicated loci for diverticular disease (connective tissue, smooth muscle and nerves of the colon), haemorrhoids, appendicitis and bowel transit. Published alleles with a protective effect were inverted to the risk direction.

Diverticular disease

Genes of the extracellular matrix (elastin, collagen VI, fibulin-3), of smooth muscle and of enteric nerve signalling.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ARHGAP15	rs4662344		T/C	imputed DR2=0.99	T (1 copy)	unfavourable (mild)	ARHGAP15 (Rho signalling of macrophages and muscle): the largest locus of diverticular disease [GWAS Catalog: Diverticular disease, allele T, OR 1.23, p=2E-18, PMID 28585551]	Strong
COLQ	rs7609897		G/T	chip	G (1 copy)	unfavourable (mild)	COLQ (neuromuscular junction collagen): diverticular disease [GWAS Catalog: Diverticulosis and diverticulitis (PheCode 562), allele G, beta 0.079 [0.063-0.096] unit increase, p=2E-23, PMID 39024449]	Moderate
SLC35F3	rs4333882		A/A	chip	G (0 copies)	favourable	SLC35F3: diverticular disease [GWAS Catalog: Diverticular disease, allele G, beta 0.007 unit increase, p=4E-22, PMID 30177863]	Moderate
GPR158	rs7086249		T/T	chip	T (2 copies)	unfavourable	GPR158: diverticular disease [GWAS Catalog: Diverticulosis and diverticulitis (PheCode 562), allele T, beta 0.077 [0.065-0.09] unit increase, p=7E-37, PMID 39024449]	Moderate
ELN	rs3757584		C/C	imputed DR2=0.96	A (0 copies)	favourable	Elastin (ELN): elasticity of the colon wall [GWAS Catalog: Diverticular disease, allele A, beta 0.120 [0.098-0.142] unit increase, p=7E-30, PMID 37492107]	Moderate
EFEMP1	rs11899380		A/G	imputed DR2=1.00	A (1 copy)	unfavourable (mild)	EFEMP1 (extracellular matrix fibulin-3): diverticular disease and hernias [GWAS Catalog: Diverticular disease, allele A, beta 0.090 [0.074-0.106] unit increase, p=4E-27, PMID 37492107]	Moderate
COL6A1	rs9983489		T/C	imputed DR2=0.97	C (1 copy)	unfavourable (mild)	Collagen VI (COL6A1): matrix of the colon wall [GWAS Catalog: Diverticular disease, allele C, beta 0.050 [0.038-0.062] unit increase, p=5E-23, PMID 37492107]	Moderate
CALCB	rs12294208		T/T	imputed DR2=1.00	C (0 copies)	favourable	CALCB (CGRP-β peptide of enteric nerves): diverticular disease and transit [GWAS Catalog: Diverticulosis and diverticulitis (PheCode 562), allele T, beta 0.077 [0.064-0.089] unit decrease; risk/increase allele = C, the opposite, p=1E-28, PMID 39024449]	Moderate

Haemorrhoids

Vascular support tissue of the anal canal (TGF- β pathway) and colonic mucus.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SMAD3	rs17293632		C/C	chip	T (0 copies)	favourable	SMAD3 (TGF- β pathway, connective tissue): haemorrhoids [GWAS Catalog: Haemorrhoidal disease, allele T, OR 1.06, p=7E-32, PMID 33888516]	Moderate
CDKN2B-AS1	rs1333047		T/T	chip	A (0 copies)	favourable	9p21 (CDKN2B-AS1): haemorrhoids (and coronary disease, in the opposite direction) [GWAS Catalog: Haemorrhoidal disease, allele A, OR 1.04, p=8E-28, PMID 33888516]	Moderate
SFMBT1	rs9847710		T/T	chip	T (2 copies)	unfavourable	SFMBT1: haemorrhoids [GWAS Catalog: Haemorrhoidal disease, allele T, OR 1.04, p=1E-23, PMID 33888516]	Weak
XKR9	rs1838392		G/G	imputed DR2=1.00	T (0 copies)	favourable	XKR9: haemorrhoids [GWAS Catalog: Haemorrhoidal disease, allele T, OR 1.05, p=5E-39, PMID 33888516]	Weak
MUC12	rs4556017		T/T	imputed DR2=0.90	T (2 copies)	unfavourable	MUC12 (colon mucus): haemorrhoids [GWAS Catalog: Haemorrhoidal disease, allele T, OR 1.06, p=1E-22, PMID 33888516]	Weak
HMGA2	rs11176001		C/C	imputed DR2=0.96	C (2 copies)	unfavourable	HMGA2: haemorrhoids and stool frequency [GWAS Catalog: Haemorrhoidal disease, allele A, OR 0.92; risk/increase allele = C, the opposite, p=2E-46, PMID 33888516]	Moderate

Appendicitis

Genes of intestinal development.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PITX2	rs2129979		G/T	imputed DR2=1.00	G (1 copy)	unfavourable (mild)	PITX2 (left-right intestinal development): appendicitis [GWAS Catalog: Appendectomy, allele G, OR 1.12, p=2E-23, PMID 30392061]	Moderate

Bowel transit and irritable bowel syndrome

Gut-brain axis and motility.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CADM2	rs1248825		C/C	chip	A (0 copies)	favourable	CADM2 (neuronal adhesion, gut-brain axis): irritable bowel syndrome [GWAS Catalog: Irritable bowel syndrome, allele A, beta 0.027 [0.02-0.033] unit increase, p=7E-15, PMID 34741163]	Moderate
BDNF	rs12273363		T/T	chip	T (2 copies)	no judgement	BDNF region: stool frequency [GWAS Catalog: Stool frequency, allele T, beta 0.032 [0.026-0.038] unit increase, p=5E-21, PMID 34957435]	Weak

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for diverticular disease, haemorrhoids, appendicitis, irritable bowel syndrome and stool frequency. Alleles are oriented towards more risk (or towards higher stool frequency) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Diverticular disease	224 (214)	248 of 428	214	97.8	unfavourable
Haemorrhoids	15 (14)	13 of 28	14	19.3	favourable
Appendicitis	9 (4)	4 of 8	4	65.4	typical
Irritable bowel syndrome	29 (26)	21 of 52	26	24.0	typical
Stool frequency	33 (31)	36 of 62	31	99.2	no judgement

6. Favourable

- **Appendicitis**: consensus at percentile **15.0** (3 genomic scores). **Moderate.**
- **Haemorrhoids**: consensus at **18.9**; loci burden at percentile 19.3; SMAD3, CDKN2B-AS1 and XKR9 without risk alleles. **Moderate.**
- **Irritable bowel syndrome**: loci burden at percentile 24.0; CADM2 without the risk allele (C/C). **Weak.**
- **Elastin (ELN)** (C/C), **CALCB** (T/T) and **SLC35F3** (A/A) without the diverticular disease risk alleles. **Moderate.**

7. Unfavourable

- **Diverticular disease:** burden of 214 loci at percentile **97.8**; score with 23,800 variants at percentile 85.5; consensus at percentile 68.7. **Moderate.**
- **GPR158** T/T (2 alleles); **ARHGAP15** T/C, **COLQ** G/T, **EFEMP1** A/G and **COL6A1** T/C with 1 allele. Effects of about 1.05–1.2× each. **Moderate.**
- **Haemorrhoids, isolated alleles:** HMGA2 C/C, SFMBT1 and MUC12 with 2 alleles; no overall effect. **Weak.**
- **PITX2** G/T: 1 appendicitis allele; no overall effect. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Extracellular Matrix Remodelling	Colon wall: EFEMP1 and COL6A1 with 1 allele, elastin without risk; diverticular disease with loci at percentile 97.8	Unfavourable (mild)
Elastin Fragmentation	Elastin in the colon wall is lost with the years and makes diverticula easier to form; ELN without the risk allele	Typical
Dysbiosis	Fibre and the microbiome protect against diverticulitis; genetically faster transit helps	Conditional
Intestinal Barrier Dysfunction	Diverticulitis and appendicitis are local failures of the barrier; appendicitis unlikely (consensus 15.0)	Favourable
Chronic Inflammation	Recurrent diverticulitis is chronic low-grade inflammation of the colon	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **High fibre every day** (already in the plan; more fibre is always better): legumes, oats, seeds, fruit and vegetables. It is the measure with the most evidence against diverticular disease and haemorrhoids. Suggestion (not applied).
2. **Avoid chronic NSAID use** (ibuprofen, naproxen): NSAIDs increase diverticulitis and its complications; CYP2C9 *1/*3 is already a reason to avoid them. Suggestion (not applied).
3. **Signs of diverticulitis**: persistent pain in the lower left abdomen with fever warrants a same-day medical assessment. Suggestion (not applied).
4. **Screening colonoscopy** as planned in the Colorectal Polyps report (colorectal risk between the 80th and 93rd percentiles in the recalculated scores): it also shows any diverticula. Suggestion (not applied).
5. **Already in the plan and protective**: daily exercise, hydration, zero alcohol and no tobacco. Nuts, seeds and corn do not increase diverticulitis. Suggestion (not applied).

10. Limitations

- For diverticular disease, the loci burden (high) and the genomic scores (typical) partly disagree; the consensus sits on the high side of typical.
- The loci for stool frequency and irritable bowel syndrome have very small effects.
- Diverticular disease, haemorrhoids and appendicitis depend heavily on diet (fibre), physical activity and medicines.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local statistical imputation (public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (12 scores: diverticular disease, haemorrhoids and appendicitis). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 5 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Maguire 2018 and Schafmayer 2019 (diverticular disease GWAS), Zheng 2021 (haemorrhoids), Orlova 2019 (appendicitis), Eijssbouts 2021 (irritable bowel syndrome), Bonfiglio 2018 (stool frequency), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Diverticular disease	ARHGAP15	rs4662344	T/C	imputed DR2=0.99	T	1	+ strand	1 copy of the risk allele
Diverticular disease	COLQ	rs7609897	G/T	chip	G	1	+ strand	1 copy of the risk allele
Diverticular disease	SLC35F3	rs4333882	A/A	chip	G	0	+ strand	without the risk allele
Diverticular disease	GPR158	rs7086249	T/T	chip	T	2	+ strand	2 copies of the risk allele
Diverticular disease	ELN	rs3757584	C/C	imputed DR2=0.96	A	0	+ strand	without the risk allele
Diverticular disease	EFEMP1	rs11899380	A/G	imputed DR2=1.00	A	1	+ strand	1 copy of the risk allele
Diverticular disease	COL6A1	rs9983489	T/C	imputed DR2=0.97	C	1	+ strand	1 copy of the risk allele
Diverticular disease	CALCB	rs12294208	T/T	imputed DR2=1.00	C	0	+ strand	without the risk allele
Haemorrhoids	SMAD3	rs17293632	C/C	chip	T	0	+ strand	without the risk allele
Haemorrhoids	CDKN2B-AS1	rs1333047	T/T	chip	A	0	+ strand (palindromic; + assumed)	without the risk allele
Haemorrhoids	SFMBT1	rs9847710	T/T	chip	T	2	+ strand	2 copies of the risk allele
Haemorrhoids	XKR9	rs1838392	G/G	imputed DR2=1.00	T	0	+ strand	without the risk allele
Haemorrhoids	MUC12	rs4556017	T/T	imputed DR2=0.90	T	2	+ strand	2 copies of the risk allele
Haemorrhoids	HMG2A	rs11176001	C/C	imputed DR2=0.96	C	2	+ strand	2 copies of the risk allele
Appendicitis	PITX2	rs2129979	G/T	imputed DR2=1.00	G	1	+ strand	1 copy of the risk allele
Bowel transit and irritable bowel syndrome	CADM2	rs1248825	C/C	chip	A	0	+ strand	without the risk allele
Bowel transit and irritable bowel syndrome	BDNF	rs12273363	T/T	chip	T	2	+ strand	2 copies of the effect allele